



Review

Fecal Microbiota Transplantation for Acute Graft-versus-Host Disease after Allogeneic Hematopoietic Cell Transplantation: Expanding the Horizon into Pediatrics

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The microbiome plays a vital role in maintaining homeostasis of the intestinal microenvironment and the immune response in allogeneic hematopoietic cell transplantation (HCT) recipients. Disruption of the intestinal microbiome has been associated with the development of acute graft-versus-host disease (GVHD) of the lower gastrointestinal tract and worse survival. Fecal microbiota transplantation (FMT) can achieve clinical responses in refractory GVHD, establishing the promise of microbiome-directed interventions in this population. Although most data on microbial changes in HCT recipients have been generated from the adult population, children with refractory GVHD represent an important group that may benefit from FMT. In this review, we first highlight characteristics that distinguish the pediatric intestinal microbiome from the adult intestinal microbiome. We then explore multiple clinical factors that warrant careful consideration to optimize the application of FMT and other microbiome-directed therapeutics to children.

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INTRODUCTION

Disruption of the intestinal microbiota is implicated in the pathophysiology of numerous health conditions, including *Clostridioides difficile* infection [1], metabolic syndrome [2], and inflammatory bowel disease (IBD) [3]. These studies have prompted investigation into microbiome-directed therapeutics as a class of interventions to treat clinical disease. The most well-established approach is fecal microbiota transplantation (FMT), a procedure in which stool from a healthy donor is transferred to the gastrointestinal (GI) tract of the recipient with the intent to ameliorate or cure disease. Initially studied in the treatment of recurrent *C. difficile* infection [4], FMT has rapidly emerged as a treatment for other conditions, including ulcerative colitis [5], diabetes [6], autism [7], and, more recently, acute graft-versus-host disease (GVHD) [8].

The use of FMT in the treatment of acute GI GVHD occurring after allogeneic hematopoietic cell transplantation (HCT) has gained momentum over the last decade. Within the HCT population, the incidence of acute GVHD ranges between 20% and 40%, with lower GI involvement associated with the

highest rate of nonrelapse mortality [9–11]. The pathogenesis of GI GVHD is traditionally believed to involve interactions between alloreactive donor T cells and damaged host tissue, but recent studies have identified the important contributions of microbial communities within the GI tract [12]. Several studies have reported an association between low gut microbiota diversity and increased mortality in adult and pediatric allogeneic HCT recipients [13–16]. Specifically, depletion of anti-inflammatory bacteria (eg, *Clostridioides*) [17–20] and mono-domination by other bacteria (eg, *Enterococcus*) [17,21] have been associated with the most severe forms of GI GVHD. In the pediatric population, children who experienced acute GVHD had significant bacterial gut dysbiosis following their first 3 to 4 months after allogeneic HCT compared to those without GVHD [22]. These findings have driven the field towards exploring FMT as an therapeutic intervention for acute GI GVHD, with early clinical reports demonstrating that FMT can reestablish microbial diversity and result in improved clinical responses [8,23,24]. Ongoing trials seek to better characterize the risks and clinical efficacy of FMT in this setting.

The bulk of the current data on microbial changes during allogeneic HCT and the use of FMT have been generated from the adult population, with pediatric studies significantly lacking. Potential differences in key clinical variables require distinction when designing and applying microbiome-directed

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interventions to children. Here we review the use of FMT in the treatment of acute GVHD and address pediatric-specific considerations, including the gut microbiota development and age-related changes, as well as factors surrounding the logistical application of FMT to this population. Given that most pertinent studies to date have been single case reports or small case series with variable dosing regimens and protocols, this clinical review provides a descriptive analysis of the available studies rather than a rigorous meta-analysis.

ACUTE LOWER GI GVHD AND FMT

Studies in the 1970 and 1980s demonstrated that selective decontamination of gut bacteria decreased the risk of GI GVHD, but in recent years, a more nuanced and sophisticated understanding of microbial communities and GI GVHD has emerged [25–27]. An association between gut dysbiosis and increased morbidity and mortality in adult and pediatric HCT recipients has been reported [12,28]. In a seminal study, Taur et al. [21] showed that adult patients with lower bacterial diversity profiles in stool at 7 days after neutrophil engraftment had a lower survival rate at 3 years compared to those with higher intestinal diversity profiles at the same time point (36% versus 67%). These findings were subsequently confirmed in an international, multicenter study by Peled et al. [14] demonstrating that lower intestinal diversity early after transplantation is associated with both transplantation-related mortality and GI GVHD-related mortality. Such results have fueled interest in applying microbiome-directed therapeutic interventions, such as FMT, to patients with GI GVHD.

First-line therapy for acute lower GI GVHD is a systemic glucocorticoid, which carries a 50% failure rate [29]. In 2019, ruxolitinib, a selective JAK1/2 inhibitor, was approved for use in adults and children age >12 years as second-line therapy for acute GVHD, with an overall response rate of 68% after 28 days and 40% at 56 days [30]. Other potent immunosuppressive agents, such as basiliximab, have been used with variable success [31]. Even with the use of multiple lines of therapy, mortality from stage 3–4 acute GI GVHD remains >50% [31,32]. Identification of clinically effective novel therapies would address a major unmet need of the transplantation community.

Tables 1 and 2 summarize the current published experience on the use of FMT to treat steroid-refractory/resistant GI GVHD in children and adults, respectively, as defined by the European Society for Blood and Marrow Transplantation-National Institutes of Health-Center for International Blood and Marrow Transplant Research Task Force [33]. Three major points should be highlighted. First, to date, only 21 pediatric patients who received FMT for GI GVHD have been included in published reports, making it difficult to provide pediatric-specific evidence to parents, providers, and investigators (Table 1). Anecdotally, many children's hospitals throughout the United States have provided FMT under a compassionate use Investigational New Drug application to the Food and Drug Administration, but those experiences typically remain largely unpublished.

Second, these studies demonstrate wide variation in protocol design, including FMT donor source, delivery route, and dosing frequency, making meta-analysis of the published experience challenging (Tables 1 and 2). Even with these challenges, it appears that approximately 50% of patients had a complete or partial response to FMT (Tables 1 and 2), which is comparable to outcomes in the REACH1 and REACH2 trials of ruxolitinib for steroid-refractory acute GVHD [30,52]. It also appears to be most effective in patients who receive FMT

shortly after diagnosis of steroid-refractory/resistant GI GVHD, suggesting that there may be a point at which the mucosal damage in the GI tract is no longer responsive to the restorative mechanisms of FMT.

Third, to date, only 2 chronic GI GVHD patients have received FMT on a study, and thus this remains an area of unexplored research [41]. Thus, at present, the number of patients receiving FMT for GI GVHD remains small. Ultimately, prospective multicenter trials will be essential to gather critical data on the safety and efficacy of FMT in pediatric GVHD patients.

AGE-RELATED MICROBIOTA CONSIDERATIONS

Establishment of a child's gut microbiota is greatly influenced by their primary source of nutrients within the first year of life and by their mode of delivery. In the United States, a longitudinal study of 107 healthy mother-infant pairs showed that breastfeeding infants received microbes such as *Bifidobacterium* and *Enterobacteriaceae* from the breastmilk and areola skin in a dose-dependent manner, which contributed to their bacterial composition even after the introduction of solid food, demonstrating that breastmilk greatly influences the establishment of an infant's microbiota [53]. In contrast, formula-fed infants are exposed to different macronutrients and micronutrients with increased levels of *Enterococcus*, *Bacteroides*, and *Clostridioides* species, thereby making the infant's microbiota more similar to an adult's [54,55].

The mode of delivery also can have an impact on the infant's gut microbiota composition. The maternal transfer of microbes differs between newborns delivered vaginally and those delivered through cesarean section. A proof-of-concept study by Korpela et al. [56] demonstrated that term newborns delivered by cesarean section can achieve a microbiota profile similar to their vaginally delivered counterparts following oral-fecal transplantation from their mother. These studies show that early nutritional and environmental exposure helps form the unique intestinal microbial composition of infants, which differs greatly from that of adults.

Building on the foundation of infancy, the microbiota of children and adolescents can vary greatly with age. Initially, it was believed that children reached their stable microbiota signature by age 3, however, it was recently shown that bacterial shifts may continue until adolescence. Roswall et al. [57] reported that Swedish children who were delivered via cesarean section experienced normalization of the microbiota between age 3 and 5 years. Although environmental or food exposures differ between the United States and Sweden, resulting in different predominant bacterial species, the microbiota age-related shifts are likely similar [58]. In contrast, Hollister et al. [59] demonstrated that the gut microbiota signatures of children age 7 to 12 years vary greatly from those of their adult counterparts primarily through the enrichment of *Bifidobacterium*, *Faecalibacterium*, and *Lachnospiraceae* species in preadolescent children.

Investigations of links between gut microbiota composition and the development of medical conditions have demonstrated an association between gut dysbiosis and childhood diseases such as IBD and, more recently, neurocognitive and gastrointestinal issues, in children with autism [7,60]. These diseases and others represent an area of expanding research, with investigators exploring the use of FMT and other microbiotas interventions with the hope of improving the quality of life of these young children.

From an age-matched exposure standpoint, one approach could be to attempt to age-match pediatric patients with FMT

Table 1
FMT Trials Including Pediatric HCT Recipients

Publication	Population	Sample Size	Pediatrics (0–18 yr)	Disease State	Lower GI GVHD Only	FMT Source	FMT Delivery and Dosing	Outcomes	AEs
Liu et al., 2022 [34]	Steroid-refractory GI GVHD	21	Yes (3 patients age 16–18 yr)	AML, ALL, SAA, MDS, HAL	Yes	Unrelated donor	Combination therapy with FMT plus ruxolitinib capsules	ORR 71% at day 28 with 10 CRs; ORR 80% at day 56; relapse rate of 33%	Viral reactivations (62%) bacterial infection (29%) and severe cytopenia (81%) were noted but unclear if related to FMT or to combination therapy (FMT vs FMT-ruxolitinib)
Merli et al., 2022 [35]	Steroid-refractory GI GVHD	1	Yes (age 5 yr)	AML	Yes	Relatives	Four NDT administrations total (re-doses when GI GVHD relapsed)	Able to wean steroids and reintroduce feeding post-FMT doses but relapsed with GI GVHD	Nausea, abdominal pain, and low-grade fever; relapsed GI GVHD weeks/months after each dose; co-administered with other GVHD therapies without SAEs
Zhao et al., 2021 [36]	Stage 4 steroid-refractory GI GVHD	41	Yes (7 patients)	ALL, AML, MDS, AA, CML	Yes	Unrelated donors	NGT or NDT	Clinical GVHD remission, median survival time, EFS and OS greater in FMT group at 90 d post-FMT	No differences in AEs between groups; mild rash, thrombocytopenia, fever, vomiting; unclear cardiac event within 3 d of FMT (cannot rule out)
Zhang et al., 2021 [37]	Steroid-refractory GI GVHD	1	Yes (age 14 yr)	MDS	Yes	Unrelated donor	Four FMT doses via NDT given weekly (first 3 doses from same donor)	Increased gut bacterial microbiota diversity.	No SAEs
Goloshchapov et al., 2021 [24]	Stage 2–4 GI GVHD, steroid-refractory	7	Yes (age 3–10 yr)	ALL, MDS, hereditary diseases	Yes	Relatives or unrelated donors	EGD, NGT, or capsules	CR (n = 3) and PR (n = 3) by 120 d post-FMT	No SAEs
Zhong et al., 2019 [38]	Steroid-refractory GI GVHD	1	Yes (age 5 yr)	WAS	Yes	Unrelated donor	Two NJT doses administered 2 d apart	CR by 3 mo with improvement seen at 15 d after FMT	No SAEs
Wong et al., 2018 [39]	Steroid-refractory GI GVHD	1	Yes (age 6 yr)	AML	Yes	Unrelated donor	NDT/NJT and colonoscopy	CR and weaned off immunosuppression	No SAEs

All patients underwent allogeneic HCT. Complete responders (CRs) defined as precipitous weaning or cessation of immunosuppression medications and resolution of GI symptoms after FMT. Partial responders (PRs) defined as partial wean of immunosuppression and improvement in GI symptoms after FMT.

AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; SAA, severe aplastic anemia; NDT, nasoduodenal tube; MDS, myelodysplastic syndrome; ORR, overall response rate; AA, aplastic anemia; CML, chronic myeloid leukemia; NGT, nasogastric tube; NJT, nasojejunal tube; EGD, endoscopy; EFS, event-free survival; OS, overall survival; SAE, severe adverse event.

Table 2
FMT Trials in Adult HCT Recipients

Publication	Population	Sample Size	Disease State	Lower GI GVHD Only	FMT Source	FMT Delivery	Outcomes	AEs
Malard et al., 2021 [40]	Grade III–IV steroid-refractory GI GVHD	76	Allogeneic HCT	Yes	Pooled healthy donor treated stool (MaaT013)	NGT (n = 2) and enema (n = 74); 1–3 doses per patient.	ORR at 28 d was 38% including 21% in CR in Phase II; compassionate use patients had a 60% ORR; higher OS in responders	5 infectious SAEs in 2 patients not found in FMT product (unable to exclude if FMT-related)
Goeser et al., 2021 [23]	Steroid-refractory GI GVHD	11	AML, MDS, thalassemia	Yes	Unrelated donors	Capsules or NJT	Significant reduction in stool output after 14 d; decreased GVHD staging	Mild nausea and abdominal pain noted
Bilinski et al., 2021 [41]	Acute and chronic GI GVHD (all stages) with antibiotic-resistant organisms in GI tract	13 (2 with chronic GVHD)	Hematologic malignancies	No, skin and liver included	Unrelated donors	NDT (3 acute patients given 2 doses, all others received 1 dose)	Decolonization of resistant organisms in 71% of patients; ORR for acute GVHD was 57%, with 42% achieving CR; 2 patients achieved CR after FMT while on ruxolitinib; all NRs died within 200 d of FMT. Both chronic GVHD patients achieved PR	High-risk cohort (ie, ECOG-PS > 2 in most patients); 2 deaths within 7 d; sepsis and septic shock < 6 h after FMT without deaths; norovirus enteritis and sepsis possibly/likely related to FMT
Bilinski et al., 2021 [42]	Stage 4 steroid-refractory GI GVHD	4	ALL, AML	Yes	Unrelated donors	NDT (at least 3 doses, up to 5)	3 in CR, 1 also with skin PR; 1 briefly responsive to FMT but improved after adding ruxolitinib	No SAEs reported
Goloshchapov et al., 2020 [43]	Stage 2–4 GI GVHD, steroid-refractory/overlap GVHD	19	Allogeneic HCT	Yes	Related and unrelated donors	EGD, NGT or capsules	Of 19 patients, 8 achieved CR and 8 achieved PR by 30 d post-FMT; only 4 PR patients achieved CR by 90 d; increased bacterial mass highest in FMT and responder groups	No SAEs
Biernat et al., 2020 [44]	Steroid-refractory GI GVHD	2	AML MDS	Yes	Unrelated donors	NGT administration (3–4 doses/patient)	CR (n = 1) and PR (n = 1); both developed liver GVHD	No SAEs
Mao et al., 2020 [45]	Steroid-refractory GI GVHD	1	MDS	Yes	Unrelated donors	FMT capsules (30 capsules/dose for 2 doses 1 d apart) plus 1 dose if relapse GVHD	GI GVHD improved but relapsed with improvement after third dose; no bacteremia	No SAEs; 1 patient died from pan-resistant <i>Klebsiella</i> sepsis not attributed to FMT (colonized)
van Lier et al., 2020 [46]	Steroid-refractory GI GVHD	15	AML, MDS, NHL, HL	Yes	Unrelated donors	One NDT administration	CR in 2/3 with resolution of diarrhea by 28 d after FMT; NR, n = 5)	No SAEs
Shouval et al., 2018 [47]	Steroid-refractory GI GVHD	7	AML, DLBCL, CML, MPN, MDS	Yes	Unrelated donor	30 frozen capsules (15 pills daily × 2)	CR 2/7 and NR 5/7; bacteremia unrelated to FMT	No SAEs
Qi et al., 2018 [48]	Steroid-refractory GI GVHD	8	ALL, AML, CML, MDS	Yes	Unrelated donors	NDT (4 with two doses, 4 with one dose)	CR (n = 4), PR (n = 2), and relapse (n = 2); those with CR were alive, those with PR or relapse died	No SAEs
Kaito et al., 2018 [49]	21-yr-old female with steroid-refractory GI GVHD	1	ALL	Yes	Relative	15 frozen capsules 4 times	Increased fecal microbiota diversity and histologic improvement after 2 FMT doses; weaned steroids after 4 FMT doses	No SAEs; increased diarrhea and hematochezia, improved after second FMT
Spindelboeck et al., 2017 [50]	Biopsy-proven GI GVHD, steroid-refractory	3	MDS, AML	Yes	Unrelated donors	Colonoscopy, 1–6 doses	Reduced stool output and immunosuppressive medications; multiple FMT doses required for engraftment	No infections related to FMT; bacteremia 10 days after FMT (n = 1); respiratory failure 9 d after FMT

(continued)

Table 2 (Continued)

Publication	Population	Sample Size	Disease State	Lower GI GVHD Only	FMT Source	FMT Delivery	Outcomes	AEs
Kakihana et al., 2016 [51]	Severe acute GI GVHD, with comorbid infections	4	AML	Yes	Relative	NDT (3 with two doses, 1 with one dose)	CR (n = 3) and PR (n = 1); safe for FMT with active comorbid infection	(n = 1); 2 deaths from septicemia at > 70 d after FMT No SAEs; for PR patient, fever 2 d after FMT with no pathogen identified; hematochezia at 4 d after second FMT dose during steroid weaning

All patients underwent allogeneic HCT. Complete responders (CRs) defined as precipitous weaning or cessation of immunosuppression medications and resolution of GI symptoms after FMT. Partial responders (PRs) defined as partial wean of immunosuppression and improvement in GI symptoms after FMT.
 NHL indicates non-Hodgkin lymphoma; HL, Hodgkin lymphoma; MPN, myeloproliferative neoplasm; DLBCL, diffuse large B cell lymphoma; EGD, endoscopy; ECOG-PS, Eastern Cooperative Oncology Group Performance Status.

donors at or around the same developmental stage. This approach would face significant logistical barriers, however, including recruitment of pediatric volunteers, determining age-matched groups (ie, infants, children, and adolescents), and maintaining a stool bank with an available supply from pediatric donors. Although age-matched donors may most closely mimic the presumed microbiota based on developmental stage, promising results have been seen with FMT products from adult donors. Moreover, it is increasingly challenging to recruit healthy adult donors and maintain an adequate FMT supply, considering that only 2.5% of donors meet the eligibility criteria [61], making targeted recruitment of age-matched FMT donors an additional barrier for children. Moreover, a study of FMT in pediatric patients with IBD showed that the response to FMT did not differ between children age < 10 years and those age > 11 years [62]. As FMT emerges as a promising therapeutic approach in GI GVHD in the adult population, pediatric GI GVHD patients should be considered for treatment with an adult FMT donor product. Age-matching FMT products to pediatric patients would create a barrier to access for pediatric patients and should not be the goal at this time.

SAFETY OF FMT ADMINISTRATION FOR ACUTE LOWER GI GVHD

Over the last 2 decades, FMT has been associated with a 19% rate of adverse events (AEs), with mostly mild AEs, including diarrhea and abdominal pain, reported. Severe AEs have been reported in just 1.4% of FMT recipients (4241 patients who received a total of 5688 FMT doses), although some events have been fatal [63]. With these safety considerations in mind, FMT remains a promising therapeutic intervention for patients with GI GVHD.

A major concern in the early experience of FMT for GVHD has been the risk of AEs, especially infections complications. Clinical trials have typically administered FMT after neutrophil engraftment to ensure the presence of neutrophils in the event of bacteremia. An early study by Taur et al. [64] showed, in a randomized controlled trial, that autologous FMT collected at 1 to 4 weeks before HCT could be safely reinfused after neutrophil engraftment and reestablish microbial diversity to previous baseline. DeFilipp et al. [65] administered donor FMT capsules to 13 patients after neutrophil engraftment and found establishment of microbial diversity and expansion of stool donor taxa. In a separate study in which FMT was administered prior to allogeneic HCT, when neutropenic on day +5 post-transplantation, 1 patient developed extended-spectrum beta-lactamase-producing *Escherichia coli* bacteremia, which was linked to the FMT donor by genomic sequencing. This case emphasizes the importance of avoiding neutropenia in FMT recipients and the need for continued evaluation and revision of donor screening protocols based on emerging infectious risks [65,66]. Subsequent reports have confirmed that FMT-related bacteremia following GVHD is likely a rare event despite the compromised integrity of the lower intestinal tract [47]. In a cohort of 47 children, 6 (13%) experienced a single episode of vomiting within 24 hours of upper GI administration of FMT for *C. difficile* infection, none of which required medical attention [67]. Finally, of the 21 pediatric patients who received FMT for GI GVHD in the present study, none experienced a severe AE (Table 1), suggesting that the safety profile is at least similar to, if not better than, that of their adult counterparts.

FMT DONOR SOURCE

Fecal source of FMT can be separated into 2 major categories: autologous FMT and traditional allogeneic donor FMT,

which can be obtained from family members or unrelated donors. The rationale for using autologous FMT is that this approach can reestablish the patient's previous intestinal microbiota while obviating the risk of transmitting foreign pathogenic organisms. As described above, autologous FMT has proven able to restore the microbiome following allogeneic HCT [64]; however, several considerations make this approach difficult to apply in the setting of GVHD. First, the majority of adult and pediatric patients undergo HCT for a malignant condition and many have recent exposure to chemotherapy, radiation, or antibiotics close to the time of HCT [68]. Those exposures have been postulated to lead to perturbations in their pre-HCT microbiota, making it a less diverse microbial product even if it is naïve to the patient [12,69]. In the aforementioned autologous FMT study, pretransplantation microbial diversity (Simpson index >2) was an eligibility requirement, which many patients in clinical practice might not achieve. Second, given the <10% incidence of severe GI GVHD after allogeneic HCT [32,70], the cost and logistics of collecting and banking stool from all patients for a rare event make autologous FMT for GVHD infeasible.

Studies and reports of donor FMT for GVHD suggest that both related and unrelated donors are acceptable sources (Tables 1 and 2). Parents of pediatric patients may prefer donation from a relative to maintain the environmental and microbiota exposures within the family. Although there may be comfort in providing a familial donor, there is also an emotional burden to the familial donor if the FMT does not “cure” the child, a point that should not be deemphasized. It is important to note that FMT donor evaluation, whether of family members or unrelated volunteers, requires weeks of rigorous workup, including an extensive medical screening and infectious disease workup, in addition to providing adequate stool collection [71,72]. Few volunteers meet the eligibility requirements for donating, and real-time evaluation of donors is too lengthy a process to provide FMT to GVHD patients on demand.

Some non-HCT studies have used pooled FMT doses to increase the availability of FMT doses. Studies in ulcerative colitis have shown that pooled FMT will increase microbial diversity and remission rates, suggesting that this approach can increase access without compromising clinical response [5,73]. For example, the Phase IIa Heracles Study (ClinicalTrials.gov identifier NCT03359980) of adult patients with acute GI GVHD treated with at least 1 dose of MaaT013, an allogeneic pooled and treated FMT product, demonstrated safe administration and improved overall survival in responding patients [40]. Although pooling FMT doses will make doses more readily available, whether donor heterogeneity will provide similar or improved outcomes as single donor FMT doses in GVHD patients is unclear. Thus, using volunteer FMT donors from stool banks offer GVHD patients the most rapid access to a healthy stool product and should be considered the current standard.

MODE OF DELIVERY

Mode of delivery represents one of the greatest barriers to FMT access in children with GVHD. FMT products can be administered as oral capsules, endoscopically (gastric, duodenal, or colonic), and by nasointestinal tube or enema. FMT traditionally has been administered in liquid form directly into the GI tract, although this approach has limitations. Patients with GI GVHD, especially those with stage 3–4 or steroid-refractory disease, represent a high-risk population for complications, such as perforation due to a friable intestinal mucosa. For many children, regardless of age, nasogastric/nasoduodenal or flexible sigmoidoscopy would need to be performed

under sedation, exposing these children to added risk. The risks of sedation and perforation may prevent parents and providers from enrolling children in FMT clinical trials. Oral capsules are an attractive approach to FMT delivery owing to ease of administration. However, most children age <8 years are not able to swallow several 00 capsules, which is required to complete a typical dose of FMT. Furthermore, children age <13 years have been excluded from receiving specific oral FMT products because of concerns surrounding the risk of aspiration and choking [74]. The ability of stool banks and companies to develop FMT products in the form of oral solutions or dissolvable tablets would greatly expand FMT access to pediatric patients, especially those whose GI tracts are compromised by physical and/or immunologic mucosal damage.

DOSING REGIMENS

There is no established FMT dosing regimen for patients with GVHD (Tables 1 and 2), and the number of reported doses ranges from 1 to 5. Dosing regimens appear to vary widely owing in part to an individual patient's response following an FMT dose. Many investigators provided additional FMT when a patient experienced a partial response or for a relapse of symptoms after initial response. In the current GI GVHD population, regardless of age, 20% of patients (48 out of 237; Tables 1 and 2) received more than 1 dose of FMT along with other immunosuppressive medications and supportive care measures to achieve symptom improvement and successful tapering of GVHD therapy. This suggests that a FMT regimen optimized to maximize clinical response likely would include multiple doses. However, the feasibility of such a regimen would be dependent on multiple factors, including the volume of FMT donor stool, route of administration, and definition of clinical response. Whether multiple planned doses of FMT may improve the duration of clinical response compared to a single-dose approach remains a major unanswered question.

Currently, there are no dosing guidelines for FMT administration in children; however, data on its use for other conditions may provide context for investigation in GVHD. Kronsman et al. [75] enrolled 10 children with recurrent *C. difficile* infections (age 1.8 to 13.6 years) and administered 30 g of resuspended donor stool (in a 100-mL suspension) as a single dose via a nasogastric tube. One patient experienced vomiting shortly after the procedure but did not require hospitalization or medical management. Nicholson et al. [76] studied 372 children (age 11 months to 23 years) who received FMT for recurrent *C. difficile* enteritis and found a remission rate of 81%, which is comparable to the adult literature [76]. The patients in this multicenter study received a median FMT volume of 240 mL (range, 120 to 250 mL), 77% via sigmoidoscopy or colonoscopy and 18% via a nasointestinal tube [76]. Given the small number of pediatric GI GVHD patients reported to date, it is difficult to recommend a dosing volume and frequency. Studies have demonstrated that FMT can be safely administered at a volume of at least 100 mL to children age <2 years in the setting of recurrent *C. difficile* enteritis, which can be used as an initial dosing strategy in Phase I trials for pediatric GI GVHD. Comparative studies will be required to determine whether an optimal standard dosing frequency for FMT exists or whether administration of repeat doses should be pursued based solely on clinical response. Separate studies will be needed in both adult and pediatric populations.

FUTURE DIRECTIONS

The availability of targeted intestinal microbiota therapies has expanded greatly in the last decade. Aside from traditional

FMT, several novel microbial restorative therapeutics have shown promising Phase III data for the treatment of *C. difficile* infection [77,78]. As promising data continue to be shared about these products, companies and investigators should pursue clinical trials that include children, possibly within a separate cohort, to examine safety and efficacy. Collaboration between investigators at multiple institutions is vital to begin to probe the immunologic mechanisms leading to these promising clinical outcomes of FMT in GI GVHD patients.

CONCLUSION

Multiple national [79] and international [63] studies have shown FMT to be both safe and effective in *C. difficile* infections and more recently in the GI GVHD population. Although the published pediatric cases represent a small cohort, it appears that FMT in the GI GVHD population has a safety profile that is at least similar to, and potentially better than, their adult counterparts. The current review provides investigators with up-to-date clinical data while highlighting pediatric-specific considerations in hopes of addressing future trial development. As the collective understanding of the role of the gut microbiota in GI GVHD continues to expand, there is an urgent need for children's hospitals to actively pursue multicenter trials to allow children access to this promising therapeutic option.

DECLARATION OF COMPETING INTEREST

There are no conflicts of interest to report.

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